Project Title:

Telco Customer Churn Prediction & Retention Insights

Project Overview

Subscription-based telecom businesses live or die by customer retention -- acquiring a new customer costs far more than keeping an existing one, yet churn is often invisible until a customer has already cancelled. The Telco Customer Churn dataset reflects the reality most retention teams face: a mix of demographic, account, and service-usage signals that, combined, hint at churn risk long before it happens.

This project aims to transform raw customer records into an actionable churn early-warning system -- surfacing which customers are at risk, why, and what a retention team could realistically do about it.

Business Objectives

- Identify customers at high risk of churning before they leave
- Understand which account and service factors most strongly drive churn
- Quantify how contract type, tenure, and billing method relate to churn risk
- Detect and correct data quality issues that could distort retention decisions
- Build a simple, interpretable churn risk classifier to support proactive outreach

Key Business Questions

1. Which customer segments churn at the highest rate?
2. How does contract type (month-to-month vs. one/two-year) relate to churn?
3. Does tenure predict loyalty, and if so, at what point does risk drop off?
4. Which services (or lack of them, e.g. no OnlineSecurity/TechSupport) correlate with higher churn?
5. Given a customer's profile, what is their estimated risk of churning?

Scope of Work

1. Data Cleaning & Preparation (SQL)

- Load raw customer records into a SQL database
- Convert TotalCharges to a proper numeric type and resolve blank entries (typically new customers with zero tenure), using CAST/COALESCE
- Standardize categorical values and check for inconsistent labeling
- Filter or correct invalid rows via WHERE-clause validation
- Document every cleaning decision -- this dataset is used precisely because its quirks mirror real customer data

2. Exploratory Data Analysis (SQL, table-based)

- Query churn rate by contract type, payment method, tenure bracket, and internet service type
- Output as aggregated tables (counts, rates, averages) rather than charts -- a deliberate choice to practice extracting insight through querying and reasoning, not visual inspection
- Segment analysis to identify which customer profiles are highest-risk

3. Churn Driver Analysis

- Rank features by their association with churn
- Identify actionable drivers (e.g. contract type, add-on services) vs. fixed attributes (e.g. demographics)

4. Churn Prediction Model (Key Deliverable)

Develop an interpretable churn risk classifier, with a personal focus on applying tree-based and boosted-tree methods in practice:

- Input: customer account, service, and billing attributes
- Output: churn risk classification (and probability), with the strongest contributing factors

Candidate models: Logistic Regression as an interpretable baseline, then Random Forest and a boosted-tree model (e.g. Gradient Boosting / XGBoost) as the primary focus -- comparing how ensembling and boosting improve on the baseline, and what each reveals through feature importance.

Deliverables

- Cleaned and well-documented dataset (via SQL)
- SQL-driven exploratory analysis of churn patterns, presented as tables
- Churn driver ranking (feature importance)
- Tuned churn prediction model, evaluated with business-relevant metrics (Recall-weighted, given the cost of missing an at-risk customer)
- Final written summary with actionable retention recommendations

Success Criteria

- Data is clean, consistent, and analysis-ready
- Insights are decision-oriented, not just descriptive
- Clear identification of the strongest churn drivers
- Model prioritizes catching likely churners (Recall) over raw accuracy, given the asymmetric cost of a missed at-risk customer vs. an unnecessary retention outreach

Business Impact

This project will enable:

- Retention teams to prioritize outreach toward genuinely high-risk customers instead of blanket campaigns
- Product/pricing teams to see which contract and billing structures correlate with loyalty
- Leadership to understand which service gaps (e.g. missing tech support) are quietly driving churn

Stretch Goal (Optional)

- Estimate potential revenue saved if top-risk customers were successfully retained
- Segment customers into risk tiers (low/medium/high) for tiered retention strategies
- Compare Logistic Regression, Random Forest, and SVM to see which best balances interpretability and performance

Final Note

The goal isn't just to predict churn -- it's to turn customer records into a retention decision-support tool a telecom business could actually act on.